

1. Administrative & Study Identification

Study Title: Safety and Efficacy Study of Fingolimod in Taiwanese Adults (\$\ge\$ 20years) With Relapsing Remitting Multiple Sclerosis **Official Full Study Title:** A 12-month, Prospective, Multi-center Post-authorization Commitment (PAC) Study Monitoring Safety in Adult Patients With Relapsing-remitting Multiple Sclerosis Newly Initiated on Gilenya (Fingolimod) in Taiwan **Short Title/Acronym:** SPRING **Posting ID:** 93877243089510820 **Protocol Number:** CFTY720DTW03 **NCT Number:** NCT04480853 **Trial Status:** RECRUITING **Sponsor / Company:** Novartis **Investigational Product:** Fingolimod (Trade Names: Gilenya, Tascenso) **ATC Code:** L04AA27 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/gilenya-epar-product-information_en.pdf **Phase of Development:** Phase 4 **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To describe the safety profile of fingolimod in the Taiwanese multiple sclerosis population, specifically focusing on the number of Adverse Events of Special Interest (AESI) during the First Dose Observation (FDO) and up to 12 months of follow-up. **Secondary Objectives:** To evaluate clinical outcomes and efficacy, including the Annualized Relapse Rate (ARR) over the study period, as well as changes in pulse and blood pressure.

2.2 Background & Rationale To fulfill a post-authorization commitment (PAC) to monitor the safety of adult patients with relapsing-remitting multiple sclerosis (RRMS) in Taiwan who are newly starting fingolimod. The study aims to collect real-world safety data for one year on patients treated based on local clinical practice, ensuring the medication's safety profile is well-documented in the Taiwanese demographic.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Post-authorization Commitment / PAC Study) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 30 patients (Note: Protocol narrative specifies up to 34 patients will be included) **Number of Sites:** 6 locations across Taiwan (including Kaohsiung,

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 20 years to 100 years. 2. Patients with a confirmed diagnosis of relapsing-remitting multiple sclerosis (RRMS). 3. Fingolimod treatment naive at the time of study entry and newly starting fingolimod based on physician judgment and according to Taiwan's fingolimod package insert (version TWI-090420).

4.2 Exclusion Criteria 1. Patients with the diagnosis of neuromyelitis optica. 2. Patients who are being treated with any investigational drug at the time of study entry. 3. In the last 6 months, experienced myocardial infarction, unstable angina, stroke, transient ischemic attack, decompensated heart failure requiring hospitalization, or Class III/IV heart failure. 4. History or presence of Mobitz Type II second-degree or third-degree atrioventricular block or sick sinus syndrome, unless patient has a functioning pacemaker. 5. A baseline QTc interval ≥ 500 msec. 6. Cardiac arrhythmias requiring anti-arrhythmic treatment with Class Ia or Class III anti-arrhythmic drugs. 7. Patient with known immune deficiency, increased risk of opportunistic infection, severe active infection, or chronic active infection. 8. Patients with severe active malignancies, except for basal cell epithelioma. 9. Patients with severe hepatic insufficiency. 10. Pregnant or nursing (lactating) women or women of childbearing potential unless on contraception.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Fingolimod 0.5 mg once daily. **Route of Administration:** Oral (capsule). **Duration of Treatment:** 12 months (1 year).

5.2 Concomitant & Prohibited Therapy Permitted: Standard multiple sclerosis treatments according to local practice. **Prohibited:** Concurrent investigational drugs, Class Ia or Class III anti-arrhythmic drugs, and immunosuppressive therapies that significantly increase the risk of opportunistic infections.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Vital Signs, Baseline QTc interval evaluation
Baseline / FDO	Day 0	First Dose Observation (FDO) monitoring for 6-hours post first dose (monitoring for bradycardia and AV block)
Interim	Months 1-11	Standard

local practice MS treatment visits, Safety monitoring (liver function, macular edema, infections), BP/Pulse checks | | **End of Study** | Month 12 | Final Vital Signs, Exit Interview, Calculation of Annualized Relapse Rate (ARR) |

(Note: If a patient requires treatment interruption necessitating a re-evaluation of FDO, the patient will be discontinued from the study. If interruption does not require FDO re-evaluation, participation continues.)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Number of Adverse Events of Special Interest (AESI). **Measurement Method:** 1. Tracked during the First Dose Observation (bradycardia and Grade ≥ 2 AV block). 2. Tracked from Baseline up to 12 months (macular edema, abnormal liver function where ALT, AST, or GGT $> 5\times$ upper normal limit, and severe lymphocytopenia where lymphocyte < 200 cells/ μL).

7.2 Secondary & Exploratory Endpoints Annualized Relapse Rate (ARR): Calculated as the total number of relapses experienced divided by the total number of days of follow-up, multiplied by 365. **Vital Signs Change:** Change from baseline of Pulse (beats/min) and Change from baseline of blood pressure (mmHg). Evaluated for notable abnormalities ($\geq 160/100$ mmHg or significant deviation from baseline).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous safety monitoring during regular clinical practice visits, with particular attention to vital signs, cardiovascular events, liver enzymes, severe lymphocytopenia, and signs of severe or opportunistic infections. **Serious Adverse Events (SAEs):** Captured throughout the 12-month study period, especially AESIs triggering treatment interruptions. **Data Monitoring Committee (DMC):** Managed by Novartis Pharmaceuticals as part of the post-authorization commitment framework.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (All enrolled patients newly initiated on fingolimod). **Sample Size Justification:** Target enrollment of $N=30$ to 34 based on post-authorization commitment requirements for the Taiwanese demographic. **Handling of Missing Data:** For patients withdrawing or switching therapies prior to 12 months, total days in study for ARR calculation is defined as the number of days from baseline to the last date in

the study. Descriptive statistics will be utilized for vital signs and AEFI frequencies.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Conducted across 6 multi-center sites in Taiwan following local clinical practice guidelines and package insert protocols. **Audit Trail:** Managed via sponsor's clinical data management systems to meet Post-authorization Commitment (PAC) regulatory standards.

11. Study Identifiers & Governance

Primary ID: CFTY720DTW03 **Registry Number:** NCT04480853 **Posting ID:** 93877243089510820 **Sponsor/Lead Organization:** Novartis **Study Title:** SPRING **Official Regulatory Title:** A 12-month, Prospective, Multi-center Post-authorization Commitment (PAC) Study Monitoring Safety in Adult Patients With Relapsing-remitting Multiple Sclerosis Newly Initiated on Gilenya (Fingolimod) in Taiwan

12. Clinical Framework (Multiple Sclerosis Specific)

Disease Area / Preferred Name: Multiple Sclerosis **Primary Condition / Preferred UMLS Name:** Relapsing Remitting Multiple Sclerosis (RRMS) **Semantic Type:** Disease or Syndrome **Definition:** A progressive autoimmune disorder affecting the central nervous system resulting in demyelination. Patients develop physical and cognitive impairments that correspond with the affected nerve fibers. **MeSH Code:** D009103 **SNOMED ID:** 24700007 **Diagnosis Threshold:** Clinically confirmed RRMS per standard diagnostic criteria, treatment naive to fingolimod **Medical Methodology:** Disease Modifying Therapy (Sphingosine-1-phosphate receptor modulator) **Optimization Target:** Reduction in Annualized Relapse Rate (ARR) and maintaining long-term safety

13. Study Procedures & Methodology

Management Pathway: Standard local clinical practice for RRMS **First Dose Observation (FDO):** Mandatory 6-hour continuous cardiovascular monitoring post-initial dose **Safety Protocol (Patient):** Routine screening for infections, macular edema, liver enzyme elevations, and lymphocytopenia **Quality Audit Mechanism:** Tracking of AEFIs and strict adherence to the Taiwan fingolimod package insert (version TWI-090420)

14. Observation & Follow-Up Schedule

Total Duration: 12 months **Onsite Visit Cadence:** Baseline FD0, with subsequent visits aligned with standard local clinical practice **Remote Monitoring Frequency:** N/A (determined by routine clinical care schedule) **Checklist Review Interval:** Continuous AE/AESI tracking up to Month 12

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Insert PI Name]	_____	07-Apr-2026					
Clinical Operations Lead	[Insert Novartis Lead Name]	_____	07-Apr-2026					
Lead Statistician	[Insert Statistician Name]	_____	07-Apr-2026					